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Best Practices in Physiologically Based Pharmacokinetic (PBPK) Modeling

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ABSTRACT

Physiologically based pharmacokinetic (PBPK) modeling is a mechanistic cornerstone of model-informed drug development (MIDD), providing evidence for regulatory decisions. PBPK models integrate physiological and pharmacological data and have been applied across a plethora of contexts of use, including assessing drug–drug interactions (DDIs), optimizing dosing for special populations (e.g., pediatrics and those with organ impairment), and supporting clinical study waivers. However, despite the established history of PBPK modeling, a significant gap persists between the quality of submitted PBPK analyses and regulatory expectations. Recent reviews by the European Medicines Agency (EMA) and other regulatory bodies indicate that many PBPK models are not deemed adequately qualified for their intended application. This discrepancy highlights the need for consolidated guidance on model development, evaluation, and application, particularly to harmonize terminology and expectations with the ICH M15 guideline. To address this challenge, this paper provides comprehensive best-practice guidance for the development, verification, validation, and applicability assessment of PBPK models. We delineate strategies for defining explicit technical criteria to ensure models are appropriately qualified and demonstrably fit-for-purpose for a specific context of use. This guidance aims to enhance the quality, consistency, and regulatory acceptance of PBPK analyses, thereby strengthening the reliability of MIDD evidence supporting drug development and approval.

1 | Background

Physiologically based pharmacokinetic (PBPK) modeling, a cornerstone of model-informed drug development (MIDD) [1], provides a mechanistic basis for understanding drug behavior by integrating physiological, pharmacological, and formulation-specific data. This framework enables the prediction of pharmacokinetics across diverse populations, species, and administration routes, thereby informing critical decisions throughout the drug development pathway [2, 3]. PBPK modeling is integral to development strategy [3] and regulatory submissions [4–6] supporting pivotal decision-making in areas such as drug–drug interaction (DDI) risk assessment

[7, 8], extrapolation to special populations (e.g., pediatrics [9–12], renal and hepatic impairment [13]), and dose selection [14, 15]. Regulatory agencies increasingly recognize PBPK modeling as a valuable tool for supporting clinical study waivers [16, 17], guiding bioequivalence assessments [18–20], and informing dosage recommendations [21] and clinical study designs [22]. In translational science, PBPK modeling bridges preclinical and clinical data, facilitating the integration of in vitro and in vivo findings to drive or support inflection point decision making and first in human dose predictions in the early stages of clinical drug development [23–25]. Consequently, PBPK applications are continuously expanding in use throughout drug discovery and development [5, 6, 26]

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and are routinely incorporated into regulatory submissions [4, 5, 27–29].

However, a recent European Medicines Agency (EMA) review of PBPK models in new medicinal product approvals indicates that regulatory expectations are frequently unmet, with many submitted models not deemed adequately qualified for their intended application [30]. The need for further guidance, best practice, and standardization was in addition highlighted during a workshop hosted by EMA in 2025 focusing on reporting and qualification of mechanistic models for regulatory assessment [31]. This observation is further supported by discussions regarding submissions to other agencies, such as the FDA [5, 6], which also highlight general challenges and knowledge gaps impeding routine utilization of PBPK models for specific regulatory decisions [32]. This underscores the imperative for best-practice documents to guide PBPK applications, with the objective of enhancing model quality, regulatory acceptance, and harmonizing terminology with the ICH M15 guideline on MIDD [33].

Despite numerous published examples and successful regulatory submissions showcasing their application, there is a notable lack of consolidated guidance on the development, evaluation, and application of PBPK models. While some high-level frameworks have been published [34, 35] and remain largely valid, they often remain too theoretical to provide clear, actionable guidance for application. Furthermore, existing literature, such as the paper by Johnson et al. [36], focuses specifically on pediatric populations. In contrast, our initiative aims to provide a comprehensive ‘good practice’ paper that standardizes the development, applications,

and documentation of PBPK modeling across a general context, ensuring utility for a broader range of applications. It details guidance for PBPK model development, verification, validation, and applicability assessment. Furthermore, it delineates strategies for defining explicit technical criteria ensuring models are appropriately qualified and demonstrably fit-for-purpose, fostering consistent and reliable MIDD evidence, supported by key regulatory guidelines [16, 37], established credibility assessment frameworks [38] and aligned with the ICH M15 draft guideline [33]. Figure 1 provides a visual overview of this best-practice framework, which is structured to follow the workflow of a PBPK modeling activity and serves as a navigational guide to the subsequent sections.

2 | Process Description and Points to Consider in Planning, Developing, Evaluating, Applying and Reporting a PBPK Model

2.1 | Planning: Understand Modeling Aim and Objectives

Robust PBPK model development requires explicitly defining the Question of Interest (QoI) and Context of Use (CoU) [16, 33, 37–40]. As established in the ICH M15 guideline, the CoU dictates the necessary model complexity, data requirements, and validation rigor [38, 40]. A prospective model risk assessment, derived from the intended model influence [33] and the consequence of a wrong decision (CoWD), determines the target level of model evaluation. To promote overall transparency and facilitate regulatory interactions and

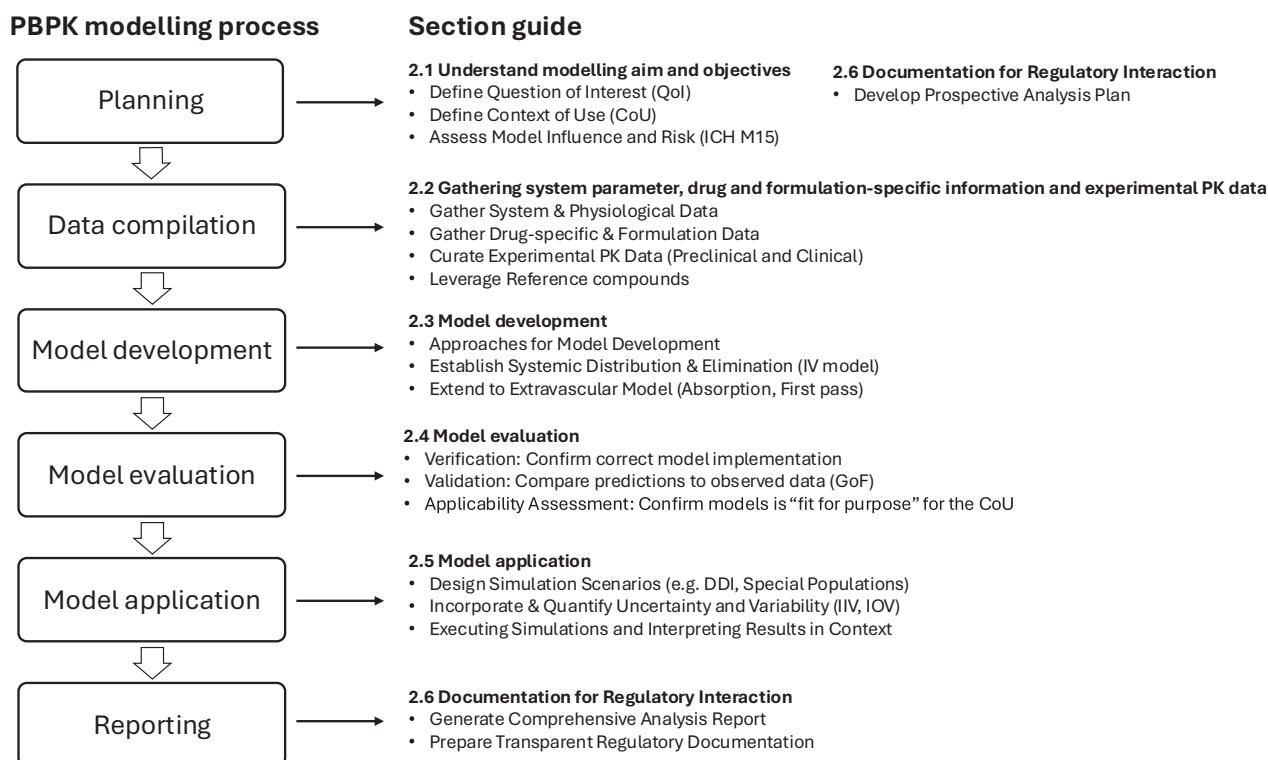


FIGURE 1 | The PBPK best-practice framework. A visual overview of the sequential process for PBPK model planning, development, evaluation, application and reporting as detailed in this manuscript.

Visualization of a general PBPK model development, evaluation and application workflow

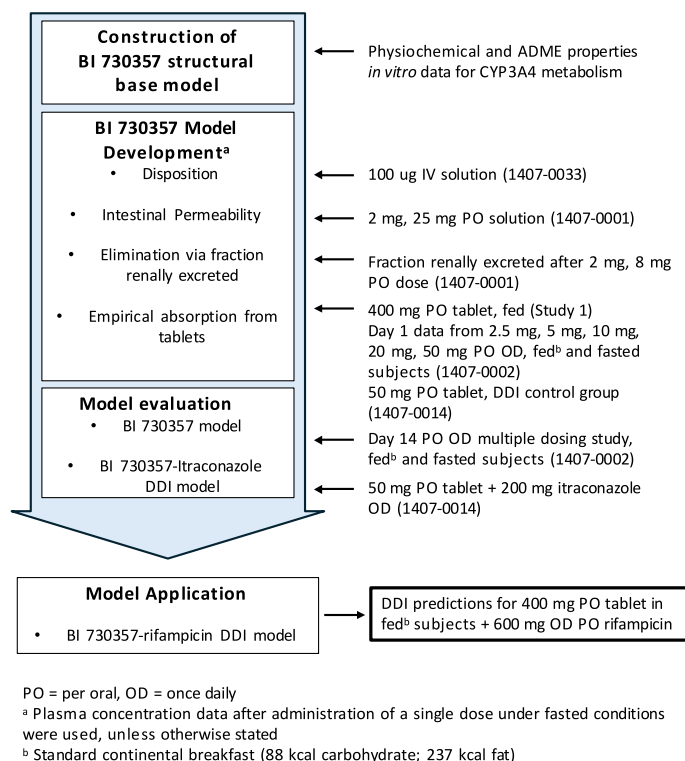
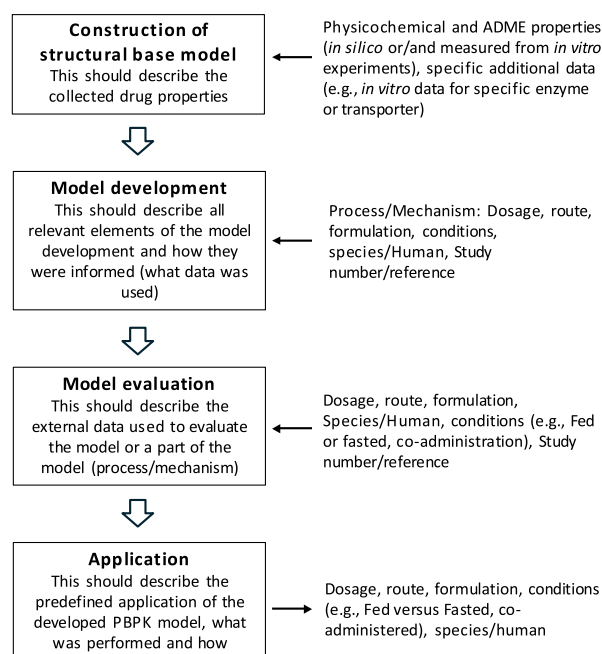


FIGURE 2 | Visualization of the PBPK modeling workflow and corresponding data sources used for model development, validation and application, and to the right a case example adopted from Kanacher et al. [44].

multi-disciplinary alignment, developers should prospectively document these elements, preferably using the ICH M15 assessment table (Table S1). While modeling objectives may iteratively evolve following a ‘learn and confirm’ paradigm as new data emerges, any modifications to the CoU must be transparently documented and justified.

The application of PBPK modeling to replace a clinical DDI study for vericiguat provides a compelling case study for the ICH M15 “assessment table.” By utilizing a PBPK model for vericiguat and its primary metabolite (M-1), its DDI liability towards the UGT1A1 inhibitor atazanavir was evaluated [41]. Although this modeling effort predated the establishment of the ICH M15 guidance and its associated framework, this retrospective analysis highlights how the vericiguat case study, which successfully replaced a clinical DDI study with PBPK-based predictions as accepted by regulators [41], can now be aligned with the “assessment table” (Table S1).

The initial definition of the modeling aim and objectives is not necessarily static, adhering to a “learn and confirm” cycle [42]. As new data become available, the understanding of the drug’s behavior may evolve. This, in turn, may necessitate refinement of the initial modeling objectives, model refinement to include emerging information, or the identification of new QoIs that can be addressed with an appropriately qualified model within a revised or expanded CoU. Any post hoc modifications to the objectives or CoU should be transparently documented and justified, particularly concerning their impact on the planned model influence.

2.2 | Data Compilation: Gathering System Parameter, Drug and Formulation-Specific Information and Experimental PK Data

Collection, curation, and critical evaluation of system parameters, drug-specific information and relevant experimental data are foundational to developing a credible PBPK model. Ideally, a comprehensive MIDD strategy identifies the necessary data, either existing or in need of generation, for model development. This cyclical process involves designing studies, potentially with existing PBPK models, analyzing the generated data to confirm hypotheses, and finally integrating these new data for model refinement [42]. The quality, precision, and completeness of these input data directly and significantly impact the predictive performance and reliability of the PBPK model [16, 37, 43]. The focus should be on the quality and information that the data brings to the model development not the quantity of data included. Regulatory authorities expect clear documentation regarding the origin, justification, and potential uncertainties associated with all data used in model development and evaluation [16, 37]. To clarify which data informs each part of the model, a schematic representation (Figure 2, modified from Kanacher et al. [44]) should be provided in addition to a data summary table. In this illustrative example, the authors predicted the impact of the CYP3A4 inducer, rifampicin, on the PK of Bevuogant/BI 730357 in fed conditions to inform future study design. Each step of the model building (e.g., disposition, intestinal permeability, renal excretion etc.) and evaluation is represented with corresponding data informing the model development.

2.2.1 | System Parameter Information

These define the physiological framework of the model. Modern PBPK software platforms include extensive, built-in databases of anatomical and physiological parameters for humans and various preclinical species, curated from peer-reviewed literature. These databases should be version-controlled and transparently sourced by the software or platform provider [45]. To simulate population pharmacokinetics and predict variability, Inter-Individual Variability (IIV) in physiological parameters is incorporated. This is often achieved by defining statistical distributions and covariance structures for key parameters within the virtual population generation algorithms of PBPK software [46]. This allows for the creation of virtual populations aimed at reflecting real-world heterogeneity [47]. However, the modeler retains the responsibility to critically assess and verify that default physiological parameters are appropriate for the specific population, disease state, and QoI under consideration.

2.2.2 | Drug Information

A thorough inventory of the investigational drug's physicochemical and in vitro ADME characteristics is essential. Key parameters include, but are not limited to, molecular weight, lipophilicity (logP or logD at physiological pH or logMA), pKa values, solubility, plasma protein binding, and intestinal permeability. Biopharmaceutics data, such as particle size distribution and dissolution profiles for orally administered drugs, are also critical. Document data generation methods (in silico, in vitro, in vivo), acknowledging inherent assumptions, variability, and limitations.

- *In silico predictions*: Often used for initial estimates (e.g., logP [48] and permeability [49]), their uncertainty should be acknowledged, and they should ideally be confirmed by experimental data where feasible. Quantitative Structure–Activity Relationship (QSAR) and Quantitative Structure–Properties Relationship (QSPR) models are widely applied computational tools for generating in silico predictions by establishing relationships between chemical structure features and biological/physicochemical endpoints [49]. Detailed discussion on these methodologies are however outside the scope of this report.
- *In vitro data*: This forms a substantial part of PBPK model parameterization. Essential in vitro ADME data include:
 - *Absorption-related parameters*: Apparent permeability from systems like Caco-2 or PAMPA, and solubility in biorelevant media.
 - *Distribution-related parameters*: Blood-to-plasma partitioning ratio, plasma protein binding (fraction unbound, $f_{u,p}$), and, if available, experimental tissue-to-plasma partition coefficients (K_p values) or data enabling their calculation (e.g., tissue composition and drug binding in tissues).
 - *Metabolism-related parameters*: For conventional small molecules, intrinsic clearance (CL_{int}) from hepatocytes, subcellular fractions, or recombinant enzyme systems, along with Michaelis–Menten constants (V_{max} , K_m) for specific enzymes. In vitro experiments can also inform

on the relative metabolizing contribution of the different enzymes, expressed as fraction metabolized (f_m), which are of crucial importance in the prediction of DDI. Biotransformation of biologics needs to be informed on a modality-by-modality basis. For example, elimination of monoclonal antibodies is primarily governed by the level of Neonatal Fc Receptor mediated salvaging from endosomal degradation, while other therapeutic peptides and proteins may be subject to proteolytic enzymes in other biophases.

- *Transporter-related parameters*: Kinetic parameters for active uptake and efflux transporters (e.g., V_{max} , K_m , or CL_{active}) derived from relevant in vitro systems, and inhibition constants (K_i) if the drug is a substrate or inhibitor of transporters. For in vitro data, document specific assay methodologies (e.g., specific cell systems such as hepatocytes, renal proximal tubule epithelial cells [RPTECs], or transfected cell lines [e.g., HEK293, MDCK], protein concentration, substrate concentration range, incubation conditions, analytical methods). If multiple sources are used to support a parameter value, the selection process and its justification should be documented. Likewise, if parameters are estimated or optimized during model development, the data and methods used should be clearly described.
- *Enzyme-perturbing parameters*: For both enzyme induction and inhibition, critical parameters for DDI assessments include EC_{50} (concentration causing 50% of maximal effect), E_{max} (maximal effect), or K_i values. Interpreting these parameters requires careful consideration of the specific in vitro assay used, as experimental setups like preincubation conditions and substrate concentrations can significantly influence the derived kinetic constants. Crucially, understanding the mechanism of inhibition (e.g., competitive, non-competitive, uncompetitive, mixed, or mechanism-based inactivation) is essential for accurate DDI prediction. For instance, a competitive inhibitor increases the apparent K_m , while a non-competitive inhibitor reduces V_{max} . Similarly, defining whether transporter inhibition is competitive or non-competitive is key for robust in vitro to in vivo extrapolation [50]. Thorough documentation of these mechanistic details, along with the methodologies, ensures the predictive power of PBPK models for complex DDI scenarios.
- *In vivo data*: Information from preclinical, and when available clinical, mass balance studies is critical for understanding the overall routes and extent of elimination (metabolism and excretion) and identifying relevant metabolites [51]. A key aspect is distinguishing between methodologies using radiolabelled compound, enabling the monitoring, identification, and quantification of the parent drug and all drug-related substances (including metabolites) by tracking total radioactivity, versus methods without radiolabel with the inherently risk of missing uncharacterized or novel metabolites that may contribute significantly to the overall drug-related material. Another critical aspect of interpreting mass balance data for PBPK modeling is accounting for conversion and reversion phenomena. This refers to the interconversion of pharmacologically active and inactive

drug enantiomers or a drug and its metabolites, or even between different metabolites, within the body (e.g., methylprednisolone and methylprednisone interconversion [52]). Radiolabelled studies, by tracking total radioactivity, help to capture these complex dynamics by providing the sum of all drug-related species. However, chromatographic and structural elucidation, often aided by enzyme incubation experiments (e.g., with glucuronidases or sulfatases), is necessary to identify and quantify individual metabolites and infer their interconversion pathways [53]. This detailed understanding of metabolite fate, including potential enteric excretion and reabsorption, is paramount for building accurate PBPK models that faithfully represent the in vivo disposition of the parent drug and its key metabolites. These insights guide the appropriate partitioning of clearance between different organs and the incorporation of relevant transport and metabolic pathways within the PBPK framework. Furthermore, integration of clinical DDI data may provide for retrospective refinement of f_m by specific enzymatic pathways. While in vitro recombinant systems and liver microsomes offer initial estimates of enzymatic contribution, these estimates may require scaling as they may not fully account for in vivo complexities such as compensatory metabolism or transporter-enzyme interplay. By leveraging clinical data from co-administration studies with potent, selective inhibitors (e.g., ketoconazole for CYP3A4 or fluconazole for CYP2C9), the in vivo f_m can be determined with higher precision. This iterative optimization ensures that the PBPK models remain predictive across diverse patient populations and polypharmacy scenarios.

2.2.3 | Formulation-Specific Information

This information is essential for biopharmaceutics application (using PBBM), where it is crucial to understand the relationship among the drug, formulation excipients, and the physiological system. Key parameters include the aqueous drug solubility at varying pH, solubility in presence of bile salts and polymers (relevant for, e.g., Amorphous Solid Dispersion (ASD) formulations), and the drug or product particle size distribution. In vitro dissolution data, ideally obtained from aqueous buffers and biorelevant media, are vital for understanding dissolution kinetics, while potential precipitation should be assessed using, for example, a two-stage assay. Additionally, critical material attributes (CMAs), like the amorphous content, should be quantified, especially when manufacturing flexibility is important. Further details can be found in the recently published PBBM report template [54].

For non-oral routes (e.g., dermal, intramuscular, subcutaneous, inhalation), specific formulation quality attributes (e.g., viscosity, droplet size, specific excipients, particle size distribution in aerosols) substantially influence both local and systemic exposure and should be similarly characterized.

2.2.4 | Experimental Pharmacokinetic (PK) Data

Observed PK data from both preclinical and clinical studies are fundamental for both the development and evaluation of PBPK

models. Complete concentration versus time profiles, including detailed absorption phase data and sufficient sampling during the elimination phase to accurately characterize clearance and volume of distribution, are of highest utility. It is imperative to thoroughly understand and document the context in which these PK data were generated:

2.2.4.1 | Population Characteristics. Comprehensive details of the study population are essential. This includes species, strain, and genetic background (for animals); for human studies, age, sex, body weight/height (BMI), ethnicity, relevant genetic polymorphisms (e.g., for metabolizing enzymes like CYPs or transporters like OATPs), health status (healthy volunteers or specific patient populations), disease state and severity (including quantitative measures of organ function, for example, liver or kidney function tests if impaired), and any co-administered medications that could be confounders. For preclinical studies, it is important to note whether studies were conducted in conscious, freely moving animals versus anesthetized animals, as anesthetics and surgical procedures can significantly alter physiological parameters (e.g., organ blood flows, cardiac output, body temperature) and thereby influence drug PK [16, 37].

2.2.4.2 | Drug Administration: Route, Dosing, Formulation and Conditions. Precise details of drug administration are required: the route of administration (e.g., intravenous (IV) (bolus or infusion), oral, transdermal, subcutaneous, inhalation, ocular), the exact dose administered (e.g., mg/kg or total dose), the dosing regimen (single dose, multiple doses with the dose interval (τ), time to reach steady state) and infusion duration for IV infusions. The exact formulation (e.g., solution, suspension, capsule, solid tablet, microarray patch, nasal spray) used in the PK study with reference to its composition if relevant (batch number can be useful for formulation prototype in early stages). For oral studies, the conditions of administration (e.g., fasted or fed state, timing relative to meals, nature and caloric/fat content of meals if fed) must be documented due to potential food effects on bioavailability [16, 37].

2.2.4.3 | Assay Methodology. The characteristics of the bioanalytical method used for drug quantification in biological matrices (e.g., plasma, blood, urine, tissue) must be well understood and reported. Key aspects include:

- *Specificity:* Whether the assay measures the parent drug exclusively, or if metabolites contribute to the measured concentration.
- *Analyte measured:* Clarity on whether the assay quantifies total drug (free plus bound) or only free drug. For studies involving radiolabelled compounds, it's critical to distinguish between total radioactivity measurements (parent + all metabolites) and specific measurements of the parent drug or individual metabolites.
- *Matrix:* The biological matrix in which concentrations were measured (e.g., plasma, serum, whole blood). If whole blood is used, the blood-to-plasma ratio is important for relating concentrations.
- *Performance:* Documented assay performance characteristics, such as accuracy, precision, linearity, and the lower

limit of quantification (LLOQ) are needed to assess the reliability of the PK data, especially at low concentrations.

- *Drug-partner complexes*: For certain drug types, like antibody-drug conjugates or drugs with high affinity, saturable binding to endogenous components (beyond typical plasma proteins), it is crucial to understand what the assay is measuring (e.g., conjugated drug, total antibody, free payload, drug-target complex).

2.2.5 | Reference Compounds

In circumstances where the PBPK model for the investigational drug incorporates complex mechanisms or is being extrapolated to new scenarios, the use of PK data from well-characterized reference compounds can be instrumental. This approach is particularly relevant when specific model extensions are made (e.g., incorporating detailed transporter kinetics, specific enzyme isoforms, or unique distribution characteristics).

These data are primarily used to evaluate the consistency and robustness of the PBPK model's description of the shared mechanisms. Successful simulation of the reference compound(s) PK, using the same mechanistic model structure and system parameters (where appropriate), lends credibility to the model's ability to characterize these specific pathways before applying it to predict the behavior of the investigational drug (or test and reference formulations in the case of virtual bioequivalence [vBE] assessment). This is a critical step in building confidence in the model, particularly for its prospective applications. If reference compounds are used:

- *Selection criteria*: These compounds should be selected based on their sharing of essential physicochemical properties and, most importantly, the same relevant physiological and biochemical ADME pathways that are being modeled for the investigational drug.
- *Data requirements*: Comprehensive and high-quality PK data for these reference compounds, under various relevant conditions (e.g., different doses, routes of administration, in the presence of inhibitors/inducers if DDI mechanisms are being evaluated), should be gathered with the same level of detail regarding population, dosing, and assay as for the primary drug.

2.3 | Model Development

Once the modeling objectives (QoI, CoU, MI) are defined and the necessary drug information and experimental data have been gathered and critically evaluated, the process of establishing the PBPK model structure commences. The development of a PBPK model is an iterative, science-driven undertaking that integrates known physiological information with drug-specific properties to simulate its PK behavior. Careful consideration of model structure, assumptions, parameterization, and data sources is fundamental to constructing a model that is fit for its CoU and capable of reliably addressing the QoI. This typically involves a cycle of model

building, parameterization, simulation, comparison with experimental data, and refinement.

2.3.1 | Navigating the Landscape of PBPK Software

The landscape of PBPK modeling is characterized by both well-established commercial software and powerful open-source platforms. Commercially available systems such as Simcyp [55, 56], with its extensive curated population and compound libraries, and GastroPlus [57], which provides detailed mechanistic models for oral absorption, have a long history of application in regulatory submissions. In 2017, the PBPK/QSP software platform consisting of PK-Sim and MoBi, which was previously available only commercially, was released as open-source software. This platform became part of the Open Systems Pharmacology (OSP) Suite [45, 58], combining complete software transparency with the methodological flexibility that characterized the earlier versions of these tools.

Regulatory authorities, including the FDA and the EMA, maintain a platform-agnostic stance given that the fundamental requirement of adequate verification is fulfilled [59]. Verification serves as the foundational quality control process to ensure that the technical implementation is both accurate and reliable. It focuses on auditing model code, confirming that mathematical equations correctly represent model assumptions, and validating the precision of all calculations. Beyond the code itself, verification encompasses the Quality Assurance of the software environment, requiring rigorous testing, installation protocols, and version tracking. To meet regulatory standards, all verification activities must be thoroughly documented and maintained for potential inspection, ensuring full transparency in how data was processed and analyzed. This ensures that the environment of the analysis is scientifically robust and reproducible, to allow for establishment of a valid PBPK model for regulatory decisions.

2.3.2 | Approaches for Model Development

The first step is to outline in what way processes should be described and informed. This is influenced by both the current level of information and the QoI. In case of mechanism-based processes, specific parameter values of each process can then either be informed purely via scaling of in vitro measurements (bottom-up approach) or optimized towards reference data (middle-out approach). In cases where processes are to be generically described, most often due to limited mechanistic knowledge, the processes can be informed solely by observed PK data (top-down approach). In complex scenarios, it is not uncommon that different approaches will be adopted for different processes depending on different levels of information. Similarly, the selected approach may need to be revised during development to achieve adequate results. Such cases should be clearly indicated and properly discussed.

2.3.3 | Establishing Systemic Distribution and Elimination (IV Model)

The foundational step in PBPK model establishment is typically the development of a model describing the drug's systemic distribution

and elimination processes (metabolism and excretion), utilizing IV pharmacokinetic data. IV administration bypasses the complexities of drug absorption, allowing the model development process to focus initially on characterizing how the drug moves between blood and tissues and how it is removed from the body. This core systemic model serves as the scaffold upon which absorption models (for extravascular routes) are later built.

2.3.3.1 | Systemic Distribution. The first stage involves selecting an appropriate mathematical description for drug distribution into various tissues and initialising model parameters based on the gathered drug information. Model structure defines the mathematical representation of the organism and the drug's disposition pathways. Key structural decisions encompass the level of detail in representing tissues and the assumptions governing drug transport across membranes.

- *Choosing between whole body versus minimal PBPK models:* A primary structural decision is the level of granularity. Whole-body PBPK models endeavor to represent all physiologically relevant organs and tissues as discrete compartments, interconnected by systemic blood flow. This approach affords a high degree of mechanistic detail, enabling prediction of drug concentrations in specific tissues and a granular understanding of disposition processes. Such models are often preferred when the QoI involves tissue-specific effects or complex distribution. Conversely, Minimal PBPK (mPBPK) [60, 61] or Lumped PBPK models [62, 63] offer a simplified representation by grouping multiple tissues with similar characteristics or combining less critical tissues into a “rest-of-body” compartment. This approach can be advantageous when available data for detailed parameterization are limited or for reducing model complexity. The selection of model structure must be justified based on the QoI, data availability, and the required level of mechanistic detail for the CoU. For instance, if the CoU is to evaluate DDI potential at the hepatic transporter/metabolizing enzyme level, a mPBPK model may still be acceptable as long as the mechanistic factors involved in the DDI process are adequately described.
- *Test different distribution models:* PBPK platforms often offer several algorithms to calculate tissue-to-plasma partition coefficients (K_p values), which determine the extent of distribution into different organs. Common approaches range from empirical correlations based on lipophilicity and plasma protein binding to more mechanistic methods incorporating tissue composition (water, lipid fractions, proteins) and the drug's physicochemical properties, including its ionization state (pKa) [64]. For example, methods described by Rodgers & Rowland, Poulin & Theil, or Schmitt and implemented in various software differ in their assumptions about binding within tissues and the influence of ionization [64]. It is important to recognize that these standard distribution algorithms function as *prediction models*, generating a priori estimates of tissue partitioning based primarily on drug physicochemical properties and average tissue compositions. The choice of distribution model should be considered carefully, justified based on the drug's properties,

and the impact of different plausible models on drug distribution predictions should be evaluated, particularly for tissues relevant to the drug's efficacy, toxicity, or elimination. Some methods explicitly use pKa values to account for differential partitioning of ionized and un-ionized species and potential ion-trapping, which can be crucial for accurately predicting distribution, especially for ionizable compounds. Regardless of which distribution method is used, the model-predicted distribution should be in agreement with the observed distribution.

- *Expected prediction performance:* Standard distribution algorithms function as *prediction models*. Given inherent biological variability and the simplifications involved in these predictive algorithms, it is generally *not expected* that a priori calculations will accurately predict tissue distribution (e.g., within a narrow margin like 20% error) across all tissues for every compound [35, 64]. Significant deviations between predicted and actual tissue distribution can occur. Modelers are encouraged to consult the original publications describing these distribution methods to understand their derivation, underlying assumptions, and reported predictive performance or limitations [64]. Comparative evaluations, such as the PhRMA CPCDC initiative assessing Vss prediction methods [65] and studies using standardized tissue compositions to compare K_p methods directly [66], have consistently shown that *no single* a priori K_p prediction method is universally superior across all chemical space (e.g., acids, bases, neutrals, zwitterions) [65, 66]. The predictive performance of a given method can vary depending on the specific properties of the drug relative to the assumptions underlying the algorithm [66]. Therefore, a recommended practice during initial model development is to evaluate the sensitivity of the PBPK model predictions (e.g., simulated PK profiles, key parameters like Vss) to the choice of K_p prediction method [66]. This may involve running initial simulations using K_p sets derived from multiple plausible methods to understand the range of potential outcomes based solely on this structural choice.
- *Permeability considerations:* The decision on the compartmental transport model should be driven by compound-specific properties which ultimately determine which process is rate-limiting. A good description of implementation can be found in the supplementary material of the paper from Fan et al. [67]
 - *Perfusion-limited models:* Assumes that drug distribution into a tissue is constrained solely by the rate of blood flow delivering the drug to that tissue, implying that cell membranes offer negligible resistance to drug passage. Simple models assume perfusion-limited distribution for highly permeable drugs.
 - *Permeability-limited models:* Are employed when the drug's ability to permeate cell membranes is the rate-limiting step in its distribution to a tissue, rather than blood flow. More complex models incorporate permeability limitations, often calculated based on drug lipophilicity and molecular size, or incorporating pKa for ionizable drugs (e.g., specific methods described by Schmitt) [64]. The choice between a reduced perfusion-limited and a more detailed permeability-limited model

should be justified based on the drug's physicochemical properties (e.g., size, charge, lipophilicity), in vitro permeability data, and known physiology of the target tissues. The assumption of perfusion- or permeability-limited distribution should be assessed for key tissues.

2.3.3.2 | Systemic Elimination. In similarity with distribution, parameters governing metabolic and/or excretory clearance are frequently adjusted to match the in vivo observed systemic clearance and PK profile shape. It is preferable to input and adjust parameters directly related to the mechanism, such as intrinsic clearance (CL_{int}) for hepatic metabolism or renal secretory clearance (CL_{sec}), rather than adjusting total plasma clearance (CL_p). These parameters represent the intrinsic capacity of the system independently of other factors, such as blood flow or protein binding, providing a more mechanistically sound basis for extrapolations. In the bottom-up approach, CL_{int} derived from in vitro systems (scaled using IVIVE—in vitro-in vivo extrapolation) provides a mechanistic basis. This involves utilizing scaling factors (e.g., microsomal protein per gram of liver (MPPGL), hepatocellularity, organ weights, blood flows) to extrapolate in vitro rates to the whole organ or systemic level [68–71]. The specific IVIVE methods, scaling factors, and underlying assumptions (e.g., well-stirred model for hepatic clearance) must be clearly described and justified. The uncertainties associated with IVIVE should be explored (e.g., via sensitivity analysis). Furthermore, data from human radiolabelled mass balance studies are highly valuable. These studies provide crucial information on the quantitative contribution of different elimination pathways (e.g., metabolism vs. renal excretion vs. biliary excretion) to the overall clearance of the drug and its metabolites. However, it is crucial that these mass balance data are critically interpreted within the context of all known or suspected disposition mechanisms for the drug. Failing to consider potential complexities, such as extensive enteric excretion and reabsorption significantly altering the apparent contribution of fecal excretion, formation of pharmacologically active or long-lived metabolites, or complex transporter interplay affecting secretion, can lead to a distorted view of the drug's overall disposition and subsequent misinformed parameterisation of clearance pathways in the PBPK model. Incorporating mass balance information allows for a more mechanistically accurate parameterisation of the respective clearance processes within the PBPK model structure, moving beyond simply fitting total systemic clearance. If multiple elimination pathways exist and mass balance data is limited or complex to interpret, information from specific in vitro phenotyping studies should be used to inform the relative contribution of each pathway, allowing for more targeted parameter adjustments (e.g., refining the CL_{int} for a specific enzyme).

2.3.3.3 | Parameter Refinement Using IV Data. If initial 'bottom-up' parameterisation does not adequately capture the observed IV PK profiles, refinement of key drug-dependent parameters is often necessary. PBPK models integrate two primary categories of parameters: those describing the biological system (organism-dependent) and those describing the drug (drug-dependent). This middle-out adjustment should be performed sensibly, focusing on parameters known to have significant uncertainty or influence on the PK profile, and always with physiological/biochemical plausibility in mind. Understanding

which secondary parameters are derived from the drug-dependant parameters are essential (e.g., the drug lipophilicity is used to calculate multiple distribution- and permeability-related parameters) as well as paying attention to the correlation of parameters in case multiple parameters are estimated simultaneously. It is often recommended to first perform manual adjustments of a few key selected parameters (identified potentially via sensitivity analysis) to understand their impact on model behavior and ensure physiological plausibility before employing automated parameter estimation algorithms. In addition to enzymatic and transporter kinetics, model refinement frequently requires the adjustments of other parameters to better capture the compound's disposition:

- **Lipophilicity ($\log P_{o:w}$ or $\log MA$):** This parameter significantly influences calculated tissue partition coefficients (K_p values) and potentially permeability. While experimentally measured or reliably predicted values should be the starting point, some adjustment might be needed to better match observed distribution volumes or terminal phase slopes. Although existing software's offer different adjustment procedures (e.g., K_p scalar or direct adjustment of the lipophilicity parameter), these procedures are basically equivalent. Adjustments should generally be kept minimal; for instance, changing an initial $\log P_{o:w}$ value by more than approximately 1 log unit often requires strong justification, especially for lipophilic compounds ($\log P > 1$). Larger adjustments might be justifiable for very hydrophilic compounds or if the initial estimate was highly uncertain. Finally, $\log D$ at physiological pH may be a more relevant descriptor for ionizable compounds as it reflects lipophilicity in biological environments.
- **Permeability:** If the default calculation (often based on lipophilicity and size) does not adequately describe the rate of distribution, permeability might be adjusted independently. This could represent refining a general permeability-surface area product (PS) or estimating specific tissue PS values. An adjustment is much context-dependent but particularly relevant if observed distribution kinetics strongly deviate from expectations, that is, slower than predicted by perfusion limitations or faster than anticipated from permeability limitations or when expecting significant contribution of membrane transporters.
- **Fraction unbound in plasma ($f_{u,p}$):** This parameter strongly influences distribution volume and clearance calculations. The unbound fraction in plasma is a critical determinant of distribution and clearance for many drugs. Accurate experimental determination is preferred. The blood-to-plasma concentration ratio (B:P) is also important for interconverting blood and plasma concentrations and clearances [47]. While it can be highly sensitive, especially for highly bound drugs (low $f_{u,p}$), adjustments should typically be constrained within the bounds of experimental uncertainty associated with its measurement.
- **f_m value:** This parameter strongly influences the impact of DDI for the victim drug. Initial in vitro f_m data can be defined using DDI study data conducted with strong inhibitor(s) of the corresponding enzyme(s). This has been illustrated in the SAR case study (Box 1).

BOX 1 | Illustrative example: refinement of f_m using clinical DDI data.

A PBPK model was developed for an SAR compound, initiating with preclinical species (rat, dog, monkey) to establish clearance, distribution, and permeability parameters. IV administration data informed the estimation of species-specific intrinsic clearance, while oral solution data served to validate intestinal permeability derived from Caco-2 cell studies. These parameters were subsequently scaled to humans by maintaining key drug properties (lipophilicity, distribution model, intestinal permeability, solubility) fixed while implementing human physiological parameters. Human clearance and Weibull dissolution parameters were estimated from Phase 1 clinical data (naïve-pooled approach; Monte-Carlo estimation). Metabolism was initially assigned based on in vitro data as 90% CYP3A4 and 10% non-CYP processes (i.e., f_m CYP3A4 = 0.9). However, this initial model build resulted in an overprediction of the observed DDI with itraconazole. Model refinement was thus necessitated. After confirming that the model adequately described the independent pharmacokinetics of the victim (SAR compound) and perpetrators (itraconazole, hydroxy-itraconazole) in a simulated population ($n = 100$), the f_m CYP3A4 was manually adjusted from 0.90 to 0.57 (with 0.43 assigned to non-CYP processes). This adjustment was performed to align model predictions with the observed clinical DDI data (e.g., AUC and C_{max} ratios), while the total intrinsic clearance value remained unchanged. The resulting refined model, incorporating an f_m value informed by clinical DDI data.

- *System parameters:* Default anatomical and physiological parameters representing the population (e.g., organ volumes, blood flows) should generally *not* be adjusted unless there is a specific, well-justified reason, such as modeling a specific disease state known to alter physiology (e.g., reduced liver blood flow in cirrhosis, altered kidney function in renal impairment) or a specific population subgroup (e.g., obese individuals). Any changes to system parameters must be clearly documented and justified.

2.3.3.4 | Systemic Distribution and Elimination Model Evaluation. The adequacy of the developed IV model is assessed by comparing its simulations against the available experimental IV PK data.

- *Data comparison:* All available relevant IV data (e.g., multiple dose levels, different infusion rates, data from different populations) should be used for model development and evaluation. Visual predictive checks (comparing simulated concentration-time profiles with observed data, using prediction intervals) and quantitative metrics (e.g., comparing simulated vs. observed AUC, C_{max} , CL, Vss) should be employed.
- *Use of IV microdose data:* In this context, IV data obtained from microdose (Phase 0) studies can also be highly valuable, particularly in early development when therapeutic dose IV data may be unavailable or difficult to obtain [72]. Such data can provide initial insights into human systemic clearance, volume of distribution, and help assess

pharmacokinetic linearity at very low exposures, thereby informing PBPK model development at an early stage [72, 73]. However, a critical consideration when using microdose data for PBPK model development is the assumption of pharmacokinetic linearity [73, 74]. Extrapolation from microdose to therapeutic dose levels may not be reliable if non-linear processes (e.g., saturable metabolism, transporter effects, target-mediated drug disposition) become significant at higher concentrations [74]. Therefore, while potentially informative, models developed or evaluated using solely microdose IV data require careful assessment regarding their applicability to the therapeutic dose range. If microdose data are used, the PBPK model may need confirmation or refinement when therapeutic dose data become available.

- A sensitivity analysis performed prior to parameter estimation is highly recommended to identify which parameters have the most significant influence on the model outputs of interest (e.g., plasma concentrations, AUC) and are therefore most suitable and necessary to include in the estimation process [33, 75]. Systematic evaluation is essential for understanding the impact of parametric uncertainty, identifying the most influential parameters that may require further refinement or data, and ultimately building confidence in the model's assumptions and its predictive capabilities [75]. It is crucial to recognize that identifying parameters associated with non-linear processes (e.g., Michaelis-Menten constants K_m and V_{max} for saturable clearance or transport) requires data that demonstrate this non-linearity, typically PK data across a range of doses where saturation becomes evident. Fitting such models to data from only a single dose level often leads to non-identifiable parameters. Bayesian methods offer a framework to integrate prior information about parameters (from in vitro data or literature, along with their uncertainty, expressed as prior distributions) with observed in vivo PK data (via a likelihood function) to derive updated parameter estimates (posterior distributions) [76]. This approach is particularly useful for sparse data situations and for robustly quantifying parameter uncertainty. Model averaging can also be employed to account for structural uncertainty [64, 65]. This involves developing multiple plausible models, each reflecting a different reasonable assumption (e.g., different K_p prediction methods or clearance scaling assumptions). By averaging their predictions, weighted by each model's probability of being correct (often based on goodness-of-fit), the resulting averaged prediction provides a more robust estimate that reflects the range of possibilities for these uncertain parameters [33, 75].
- *Value of IV data and alternatives:* IV data are highly valuable because they allow distribution and elimination processes to be characterized largely independently of absorption complexities. Establishing a robust IV model provides a strong foundation for subsequent development of models for extravascular routes. Developing a PBPK model using only per oral (PO) data is possible but introduces significant challenges and uncertainties. Without IV data, estimates of systemic clearance (CL) and volume of distribution (V) are confounded by the unknown bioavailability (F) and absorption rate (k_a). The model must then rely more

heavily on IVIVE predictions for clearance and PBPK methods for distribution, potentially increasing uncertainty. An alternative option is to fix the absorption (permeability) to measured or predicted values and still estimate clearance. This uncertainty is particularly pronounced for drugs with low solubility and/or low permeability (BCS Class II, IV), or those subject to significant gut-wall metabolism or transporter effects, where absorption processes heavily influence the observed PK. In situations where human IV data are unavailable, PK data in humans from an oral solution or immediate-release (IR) formulation may be a better alternative to animal data when the drug shows complete gut absorption and/or has >85% absolute bioavailability. In some instance, if animal IV and PO data exist, scaling an animal PBPK model (validated with both IV and PO data) to humans might be a viable strategy. If IV data are completely lacking (both human and relevant animal) and the drug does not show complete gut absorption nor bioavailability > 85%, the suitability and credibility of using a PBPK approach must be carefully evaluated in light of the specific QoI and CoU. The increased uncertainty associated with key systemic parameters (CL, V, F) must be acknowledged and assessed (e.g., through extensive sensitivity analysis) to determine if the model can still provide sufficiently reliable information for its intended purpose [35].

- *Initial parameter adjustments and need for optimization:* Before adding more complex mechanisms (like active transport), it is essential to evaluate the model's performance using the chosen distribution and permeability approaches and the initial 'bottom-up' parameters according to the middle-out approach. As highlighted by comparative initiatives [65], while a priori tissue-composition based K_p predictions provide necessary initial estimates for PBPK models, subsequent optimization or refinement of tissue distribution parameters is often necessary and expected once relevant in vivo reference data (e.g., observed plasma PK profiles suggesting specific apparent volumes of distribution or distribution kinetics, or ideally, measured tissue concentration data) become available [65]. Adjustments to uncertain drug-dependent parameters within scientifically plausible ranges should be explored at this stage to assess the basic model's capability to describe the observed IV PK data, serving as a bridge between initial prediction and data-informed refinement using in vivo evidence.

2.3.3.5 | Model Structure Refinement. If adjusting plausible parameter ranges within the chosen model structure fails to adequately describe the IV data (e.g., cannot capture observed concentration levels, profile shape, or dose non-linearity), it may indicate that relevant physiological or biochemical processes are missing from the model.

- *Rationale:* Discrepancies between simulation and observation suggest the need to incorporate additional mechanisms known or hypothesized to influence the drug's PK.
- *Adding processes:* Based on prior knowledge (in vitro data, literature, pharmacological class effects) or as a testable hypothesis derived from the data trends, additional processes might be incorporated. Common examples include:

- *Active transport:* If active transport significantly influences ADME, in vitro-derived kinetic parameters such as K_m , V_{max} , or CL active for relevant transporters (e.g., P-gp, BCRP, OATPs, OCTs) are required [47]. Adding carrier-mediated transport (influx or efflux) across sinusoidal or canalicular membranes in the liver, renal tubules, blood-brain barrier, or other tissues. Careful attention must be paid to the directionality of transport (uptake vs. efflux) and the specific membrane involved. Advanced approaches, such as proteomics-informed PBPK modeling, are emerging to provide more quantitative and mechanistic insights into transporter abundance and activity, thereby improving the prediction of drug-transporter interactions and tissue-specific exposure [77].
- *Specific binding:* Incorporating saturable binding to plasma proteins (beyond albumin/AAG), tissue components, or pharmacological targets, especially if high affinity binding (K_d) and limited binding capacity are suspected and could explain non-linear distribution.
- *Enterohepatic recirculation:* If secondary peaks are observed in PK profiles after IV administration, biliary excretion followed by reabsorption might need to be included.
- *Parameterisation:* When adding new processes, parameters governing them (e.g., transporter K_m , V_{max} ; binding K_d , biliary clearance rate) should ideally be informed by experimental data. If such data are unavailable, parameters may need to be estimated by fitting the model to the in vivo PK data, but this requires caution regarding identifiability and the estimates should be assessed for biological plausibility.

2.3.4 | Extravascular Model: Characterizing Absorption and First-Pass Effects

Once a robust PBPK model describing the systemic distribution and elimination characteristics of the drug has been established and verified using IV data (the "IV model"), the model can be extended to describe PK profiles following PO or other extravascular routes of administration. A cardinal principle in this sequential approach is that parameters primarily influencing systemic distribution and elimination (e.g., tissue partition coefficients, systemic clearances), which were defined and validated with IV data, should generally remain fixed or be subject to minimal, highly justified changes during the development of the PO model. The primary focus of the PO model development is to accurately characterize the processes governing drug absorption from the gastrointestinal (GI) tract and any first-pass extraction in the gut wall and liver.

2.3.4.1 | Adjust Parameters Influencing Absorption. In analogy with IV model development, if initial 'bottom-up' parameterisation does not adequately capture the observed PO PK profiles, refinement of key PO

drug-dependent parameters is often necessary. When fitting the model to PO data, adjustments will typically then focus on parameters directly related to the rate and extent of drug absorption:

- *Intestinal permeability (P_{int}):* Initial estimates of P_{int} are often derived from in vitro measurements, for example, cell monolayer or artificial membrane assays, or in silico predictions based on physicochemical properties. These a priori permeability values need to be appropriately scaled given the parameterization of the absorption model at hand. An adjustment by a factor of up to approximately 3-fold is often considered reasonable to describe the observed fraction absorbed and absorption rate for BCS Class I drugs. For drugs with predicted permeability and/or solubility limitations, larger adjustment factors might be justifiable, but should be supported by an understanding of the drug's properties and potential limitations of the initial prediction method. In addition, the potential for colonic absorption needs to be assessed.
- *Solubility in gastrointestinal fluids:* The drug's solubility in the various regions of the GI tract is a critical determinant of its dissolution rate and extent of absorption, especially for poorly soluble compounds (BCS Class II and IV) [47]. Ideally, solubility data from biorelevant media (e.g., Fasted State Simulated Intestinal Fluid (FaSSIF), Fed State Simulated Intestinal Fluid (FeSSIF), Fasted State Simulated Gastric Fluid (FaSSGF)) should be used as input parameters [47]. If only aqueous solubility data were initially used, adjustment of solubility parameters during PO model development is often justified and necessary to accurately reflect the in vivo dissolution environment, particularly when modeling fed versus fasted conditions.
- *Formulation-dependent parameters:* For solid oral dosage forms, parameters describing the drug release characteristics from the formulation are key inputs. These include but are not limited to: dissolution rate constants (which may be derived from in vitro dissolution testing under various conditions), disintegration times, and lag times for drug release (e.g., for enteric-coated or delayed-release formulations). These parameters are specific to the formulation administered in the clinical study and are primary candidates for adjustment when fitting data from different formulations of the same drug.
- *Parameters of meal events (food effects):* Co-administration of food can significantly alter drug absorption through various physiological changes. Standard PBPK models include parameters to simulate these alterations, such as modifications to gastric emptying rate, intestinal transit times, GI fluid volume and composition (pH, buffer capacity, osmolality), splanchnic blood flow, and importantly, the secretion and concentration of bile salts [78, 79]. When fitting PK data from fed-state studies, these physiological parameters within the PBPK model, or the drug's sensitivity to them (e.g., impact of bile salts on solubility and dissolution for lipophilic drugs), may be adjusted to capture the observed food effect on the drug's PK profile [78, 80].
 - *Physiological response dependency to food constituents:* The magnitude and nature of these GI physiological

changes are often dependent on the composition and caloric content of the ingested meal [79, 81]. For example: gastric emptying is generally delayed more by high-fat/high-calorie meals; gastric pH typically increases post-meal, influenced by food buffering capacity; bile salt secretion, crucial for lipophilic drug solubilization, is markedly stimulated by fats; and splanchnic blood flow can increase, particularly with carbohydrate/protein-rich meals [78, 79, 81].

- *Modeling different meal types:* When developing PBPK models to predict or explain food effects, it is therefore important to consider the type of meal administered in clinical studies (e.g., standard FDA high-fat, high-calorie breakfast vs. low-fat meal). Advanced PBPK absorption models allow for adjustment of GI physiological parameters to reflect these different meal compositions [78, 79]. If the aim is to predict the effect of various meal types, the PBPK model needs to be parameterized and validated accordingly, ideally using clinical data from studies employing those distinct meal conditions.

2.3.4.2 | Modify Model Structure for Oral Absorption. If adjustments to the absorption parameters are insufficient to describe the observed PO PK profiles, or if known mechanisms influencing oral bioavailability are not yet included, structural modifications to the GI tract model may be necessary. This often involves incorporating:

- *Intestinal active transport processes:* If the drug is a known substrate for intestinal influx (e.g., PEPT1, OATPs) or efflux transporters (e.g., P-glycoprotein (P-gp), BCRP), these processes should be incorporated into the intestinal segments of the PBPK model [47]. This requires parameterization of transporter kinetics (e.g., K_m , V_{max}), which should ideally be informed by in vitro data; however, developing reliable, quantitative, biopredictive assays for most transporters remains challenging. The directionality (uptake from lumen or efflux into lumen) and location (apical or basolateral membrane of enterocytes) of these transporters are critical.
- *Gut-wall metabolism (first-pass intestinal metabolism):* If the drug undergoes significant metabolism in the enterocytes (e.g., by CYP3A4), this first-pass effect must be considered. This typically involves specification of local enzyme abundances within the enterocyte compartment. Data from in vitro studies with intestinal microsomes or S9 fractions, or specific human mass balance data differentiating intestinal from hepatic metabolism, can inform these parameters.
- *Other GI processes:* Depending on the drug, other processes like luminal degradation, re-conversion of metabolites, or supersaturation and precipitation dynamics might need to be considered and structurally incorporated. Any structural modification should be mechanistically plausible, supported by available in vitro or preclinical evidence, and its impact on model predictions thoroughly evaluated.

2.3.4.3 | Comparing Simulations With Experimental PK Data for Oral Administration. The principles for evaluating the PO model against experimental data are similar

to those for the IV model, involving visual predictive checks and quantitative comparison of PK parameters (AUC, $C_{\max}$, $T_{\max}$).

- *Specific strategy for different formulations:* A recommended best practice, if PK data are available for both an IR formulation and one or more modified-release (MR) or controlled-release (CR) formulations, is to adopt a stepwise approach:
 - First, establish and validate the absorption model using data from the IR formulation. During this step, parameters like intestinal permeability, solubilities, and potentially gut-wall metabolism/transport parameters are adjusted. This step should preferably also include data from an oral solution or suspension to delineate any influence from the IR formulation.
 - Once the IR model adequately describes the data, this core absorption model (with its now-defined parameters) is used as the basis for modeling the MR/CR formulations. For the MR/CR data, only the formulation-specific drug release parameters (e.g., dissolution profiles, release rate constants) should be adjusted. Parameters related to intrinsic permeability, solubility (unless formulation significantly alters it in a way not captured by release), and systemic disposition should remain fixed from the previous steps. This approach helps to deconvolve drug-specific absorption properties from formulation-specific release properties. Still, if absorption extends into colonic segments for MR/CR formulations, regional intestinal permeability may require reassessment and cannot always be directly informed from the IR model.

2.3.5 | Note on Sequential Versus Simultaneous IV/PO Model Development

The stepwise approach described (establishing the IV model first, then fixing systemic parameters and developing the PO absorption model) is generally recommended. This ensures that information from IV data, which directly informs systemic distribution and clearance processes without the confounding factor of absorption, is appropriately leveraged. This separation helps to improve parameter identifiability for systemic versus absorption-related parameters. However, there are situations where this strictly consecutive model building might not be optimal, and a simultaneous fitting approach using both IV and PO data might be preferred or necessary [35]:

- *Shared mechanistic pathways:* If the PBPK model incorporates processes that are critical for both systemic disposition after IV administration and oral absorption/first-pass effects (e.g., significant hepatic uptake transport influencing both systemic clearance and hepatic first-pass extraction; saturable systemic metabolism that also contributes to first-pass metabolism), then simultaneously fitting IV and PO data can provide more robust and identifiable estimates for the parameters governing these shared pathways.

- *Data richness and quality:* If the available IV data are sparse (e.g., limited sampling times, short duration profiles, high variability) while the PO data are more extensive and informative, a simultaneous fit might better leverage the richer dataset to inform shared parameters, provided the model structure is appropriate. Appropriate weighting of the different datasets (IV vs. PO, different dose levels) is crucial in such simultaneous estimations.
- *Specific modeling objectives:* If the primary QoI heavily relies on accurately characterizing processes like first-pass extraction that are only observable after PO administration but are mechanistically linked to systemic parameters, a simultaneous approach might be more direct. Even when a simultaneous approach is chosen, the fundamental principle of trying to distinguish systemic disposition from absorption processes as much as possible should guide the modeling strategy and parameterisation. Sensitivity analyses and careful evaluation of parameter correlations are particularly important in simultaneous fitting to ensure robust parameter estimation [75].

2.3.6 | Other Extravascular Routes

While oral administration is the most common route, PBPK models are increasingly developed and applied to understand and predict drug disposition following various other extravascular routes, including pulmonary (inhalation) [82], dermal [83], subcutaneous [84], intramuscular, ocular [85], and nasal delivery [86]. Detailed modeling strategies and data requirements are highly specific to each administration route. For instance, PBPK modeling of pulmonary delivery necessitates careful consideration of particle deposition models, dissolution in lung lining fluid, mucociliary clearance mechanisms, and absorption across the alveolar-capillary barrier [87]. Similarly, models for subcutaneous administration must account for factors such as local blood flow and lymphatic drainage at the injection site, drug binding to interstitial components, potential local degradation, and the influence of injection volume and formulation properties on absorption kinetics [88]. PBPK models for dermal absorption often include multiple skin layers (stratum corneum, viable epidermis, dermis) with distinct diffusional and partitioning properties [89]. Developing credible PBPK models for these routes requires a deep understanding of the route-specific physiology, formulation science, and drug properties, along with relevant experimental data to inform and validate the model structure and parameters.

2.3.7 | Note on Parameter Estimation

If parameters (e.g., refined CL_{int} , transporter kinetics, binding parameters) need to be determined by fitting the model to the data, formal parameter estimation (optimization) techniques are used. This process aims to find parameter values that minimize the difference between model predictions and observed data according to a defined objective function. Any parameter estimation or optimization strategy should be prospectively defined, with clear justification for the parameters selected for

optimization, the data used, the objective function, and the method employed. The resulting parameter estimates should be critically evaluated for their mechanistic plausibility. When estimating multiple parameters simultaneously or when data are sparse, assessing parameter identifiability is crucial. Approaches to address non-identifiability include stepwise parameter estimation, incorporating informative prior distributions (Bayesian approaches), or conducting local and global sensitivity analyses to ensure that the data contain sufficient information to uniquely determine the parameters.

2.3.8 | Note on Sensitivity Analysis and Parameter Uncertainty Quantification

Local sensitivity analysis, conducted using a base case and perturbing parameters one-at-a-time, is used to identify the primary drivers influencing the model outcomes and highlighting any non-linear “tipping points” where the output is most volatile. Such analyses are readily reported via Tornado plots for ranking variable impacts and Spider plots for visualizing linearity and rates of change. Global sensitivity analysis explores the entire parameter space simultaneously to quantify how interactions, correlations, and collective variations drive model uncertainty, often visualized through Sobol indices or scatterplots. Report these findings by highlighting total-effect indices to account for multi-variable synergies, providing a more comprehensive risk assessment than isolated, one-at-a-time perturbations [75, 90–93].

While global sensitivity analysis and Monte-Carlo simulation-based uncertainty propagation methods are applied to quantify the relative contribution of each process to the total uncertainty of the model outcome, Maximum Likelihood Estimation and advanced Bayesian methods have been proposed for quantification of population parameters, variability, and uncertainties. A Bayesian Hierarchical Meta-Analysis approach has also been suggested to assess bias and imprecision on a platform level by analyzing collated data from multiple reference studies. Still, it is currently premature to establish definitive best practices for uncertainty quantification, and it is important to remember that the output from conventional PBPK models does not intend to fully replicate clinical trial outcomes as residual uncertainty is not an integrated part of the fundamental methodology.

2.4 | Model Evaluation

The primary objective of model evaluation is to establish sufficient credibility that the model is fit-for-purpose within its defined CoU [16, 33, 37]. The required evaluation rigor must be proportional to the model risk [33, 94]. Evaluation is not a single event but an iterative process, and should be integrated throughout model development and refinement, where findings inform further model improvements until the necessary level of credibility for the CoU is achieved [16, 33, 40]. Aligned with these principles, PBPK model evaluation encompasses: (1) Verification: confirming the mathematical and computational integrity of the model implementation; (2) Validation and applicability assessment: demonstrating the model's accuracy in representing

the biological system through qualitative and quantitative comparisons against experimental data and ensuring the validated model scope, data, and underlying assumptions are adequate to address the specific QoI.

2.4.1 | Model Verification

Model verification focuses on ensuring the integrity and correctness of the PBPK model implementation [33]. This involves confirming:

- *Correctness of implementation:* The PBPK model equations representing physiological processes, drug-specific properties, and any structural assumptions are correctly translated into the modeling software code. This may involve code review, comparison with standard model implementations, or unit testing of individual model components.
- *Mathematical soundness and algorithmic performance:* The underlying mathematical equations are appropriate and solved accurately by the software's numerical algorithms. This can involve checking for mass balance, assessing sensitivity to solver settings (e.g., step size, tolerance), and ensuring solution convergence.
- *Validated computerized system and quality assurance:* Ensuring that modeling activities are reliable, reproducible, and traceable and that adequate quality assurance is performed for data management and modeling activities.

It should be noted that in applications using predefined PBPK platforms, software-level verification of the mathematical implementations and parameterization is often conducted by the provider of the platform. Ideally, these verification processes should be transparent and accessible for peer review; otherwise, full accountability rests with the provider. Nevertheless, the onus is always on the modeler to confirm that the model architecture is appropriate for the QoI and that it is applied correctly within the study.

2.4.2 | Model Validation

Model validation aims to establish that the PBPK model provides a credible and accurate representation of the relevant aspects of the biological system for the defined CoU [33]. This is primarily achieved by comparing model predictions with experimental data:

- *Goodness-of-fit assessment:* This involves both qualitative and quantitative comparisons:
 - *Graphical evaluation:* Visual inspection of overlay plots comparing simulated concentration-time profiles (for plasma, and tissues/excreta if available and relevant) against observed experimental data, including measures of central tendency and variability (e.g., mean profiles, individual data points, prediction intervals) [16, 37, 95]. Plots of observed versus predicted values for concentrations and key PK parameters (e.g., $C_{\max}$, AUC, $t_{1/2}$), and various types of residual plots (e.g., weighted residuals vs. time or prediction) should be used to identify systematic deviations, biases, or trends [95].

- *Quantitative metrics*: Calculation of statistical metrics to objectively assess the agreement between predictions and observations. These should include measures of overall agreement, bias, and precision. Common metrics include prediction error (PE), average fold error (AFE), absolute average fold error (AAFE), and root mean squared error (RMSE) for PK parameters and/or concentrations. For instance, AFE and AAFE are preferred for assessing systematic bias and average error, while RMSE is useful for overall predictive performance. Acceptance criteria, set with the CoU and model risk in mind, should be prospectively predefined in the analysis plan (e.g., predictions for key exposure parameters like AUC and $C_{\max}$ should be within a predefined fold range of observed values). The 0.8–1.25-fold criterion is typically appropriate for virtual bioequivalence (vBE) assessments [96], whereas a more flexible 2-fold criterion may be justified for early-stage development.

While the aforementioned metrics provide a foundational assessment of model performance, several alternative expressions and refinements can offer more nuanced insights or facilitate easier interpretation across different datasets. Many of these measures represent mathematical transformations of the primary metrics; for instance, while AFE provides a measure of bias, the Mean Logarithmic Error (MLE) and Geometric Mean Fold Error (GMFE) offer standardized ways to express this central tendency on a logarithmic scale, preventing large values from disproportionately influencing the bias assessment. Similarly, refinements of precision such as the Relative Root Mean Squared Error (rRMSE) normalize the RMSE to the observed mean, allowing for a more direct comparison of model performance across different dose levels or physiological compartments. Furthermore, metrics such as the Concordance Correlation Coefficient (CCC) and Adjusted *R*-squared extend the analysis of ‘overall agreement’ by accounting for repeated measures or penalizing for model complexity, ensuring that the assessment of predictive power remains robust against overfitting.

- *Internal and external validation*:

From a strict best-practice perspective, a PBPK model is considered fully validated only after both internal and external evaluations have been successfully demonstrated. Models relying solely on internal evaluation due to a lack of independent data carry evidentiary limitations. These must be clearly communicated as part of the residual uncertainty and should inform the weight assigned to model outputs in decision-making.

- *Internal validation*: Assesses how well the model describes the data used for its development and parameter estimation.
- *External validation*: Provides a more rigorous test of predictive performance by comparing model predictions against independent data sets that were not used during model development or calibration. Using relevant but independent data, this increases confidence for the proposed application and demonstrates the model’s generalizability [33].

- *Consideration of uncertainty*: It is crucial to acknowledge and quantify uncertainty in both model predictions (stemming from structural and parameter value uncertainties) and experimental data (arising from biological and analytical variability) to ensure a scientifically sound comparison [16, 33, 37, 97]. While the initial application of Bayesian approaches has demonstrated the potential for joint uncertainty estimation [98], additional studies focusing on their robustness and generalizability are necessary to inform wider acceptance and implementation of this method.

2.4.3 | Applicability Assessment

Following verification and validation, an applicability (or “fit-for-purpose”) assessment should be conducted to determine if the evaluated model is indeed suitable and adequate to address the specific QoI within the defined CoU [33]. This involves considering:

- *Alignment of model scope with QoI and CoU*: Does the model include all key physiological and drug-specific mechanisms relevant to the question being addressed? Are the model assumptions appropriate and justified for the specific application and target population?
- *Adequacy of validation for intended use*: Is the validation evidence (e.g., range of doses, populations, conditions under which the model showed good performance) relevant and sufficient for the intended prospective application of the model? For example, a model validated only for single-dose predictions in healthy adults may not be directly applicable for predicting multiple-dose PK in a pediatric population without further justification or adaptation
- *Limitations and assumptions*: A clear understanding and articulation of the model’s limitations, underlying assumptions, and the domain within which its predictions are considered reliable are essential parts of the applicability assessment. The assumption table proposed in the Best practices paper in Model-Informed Drug Discovery and Development, with assumptions and justification, can be used for this purpose [99].
- *Robustness to uncertainty*: The impact of uncertainties (identified through sensitivity analyses, scenario testing, and uncertainty propagation) on the model’s robustness for the specific QoI needs to be evaluated. If model conclusions are highly sensitive to uncertain parameters that cannot be further refined, this needs to be considered in light of the consequences of wrong decisions for the patient and may limit the model influence in the decision making [33, 97].

2.5 | Model Application

Once a PBPK model has been adequately developed, verified, validated, and applicability assessed for its predefined CoU, it can be applied to address the specific QoI. The application phase involves designing and executing simulation studies that leverage the mechanistic framework of the PBPK model to generate evidence supporting drug development decisions or regulatory

submissions. This process should be prospectively outlined in the analysis plan and its outcomes thoroughly documented in the analysis report [33].

2.5.1 | Designing Simulation Scenarios Aligned With the Question of Interest

The design of simulation studies should be directly guided by the QoI and the specifics of the CoU.

- *Defining virtual populations*: Simulations are typically conducted in virtual populations that accurately represent the target patient population(s) for which predictions are sought. This includes defining subject characteristics such as age, sex, ethnicity, body weight distribution, and relevant genetic polymorphisms (e.g., for metabolizing enzymes or transporters if pertinent to the QoI). PBPK models are powerful tools for predicting PK in populations where clinical studies may be challenging to conduct. The model structure generally remains consistent, but the underlying system parameters (physiology) are adjusted to reflect the specific population:
 - *Pediatrics*: Requires incorporation of age-dependent changes in anatomical and physiological parameters (e.g., organ volumes and blood flows relative to body size/age, body composition) and, crucially, the maturation profiles (ontogeny functions) of key metabolic enzymes (e.g., CYPs, UGTs) and transporters [36]. Platforms like the OSP Suite incorporate extensive libraries of pediatric physiology and ontogeny functions [47].
 - *Geriatrics*: Must account for age-related physiological alterations such as potential declines in renal function (GFR), changes in hepatic blood flow and enzyme activity, altered body composition (e.g., increased fat mass, decreased lean mass), and differences in plasma protein concentrations [100, 101].
 - *Disease states (e.g., organ impairment, obesity)*: Model parameters are modified to reflect the pathophysiological impact of the disease on ADME processes. Examples include adjusting GFR and potentially transporter expression for renal impairment [102]; modifying liver volume, blood flow, enzyme activity, and protein binding for hepatic impairment; or altering body composition parameters, organ sizes, and blood flows for obesity [103, 104].
 - *Pregnancy and breastfeeding*: Requires dynamic adjustments for physiological changes throughout gestation and during breastfeeding, including altered organ volumes, blood flows, plasma protein binding, and enzyme/transporter activity. PBPK models, often within platforms like the OSP Suite, can integrate these complex changes to predict drug disposition in pregnant women and breastfeeding women, as well as assess fetal exposure [105, 106]. The justification for these physiological modifications must be based on literature data and clearly documented
- *Considerations for biopharmaceutics applications*: In the context of PBPK modeling for biopharmaceutics applications (also referred to as PBBM), early decisions regarding dissolution modeling are essential. These PBBM approaches are key components of broader, evolving strategies for in vitro–in vivo correlation/relationship (IVIVC/R) models which are crucial for accelerating development, reducing costly experimental studies and clinical trials, and for justifying regulatory decisions for drug formulation related QoI [107]. Of note, a recently published PBBM report template summarizes model considerations and proposes a structure for presenting QoI, model context, input data, a modeling plan, and validation [54]. Key considerations include:
 - *Mechanistic versus empirical modeling of dissolution*: Deciding whether to use a mechanistic approach to model dissolution kinetics, which may provide a more accurate representation of the drug product's release profile under physiological conditions, or an empirical approach that relies solely on observed PK data [108].
 - *Linking in vitro dissolution to in vivo dissolution (PBPK-based IVIVC)*: Assessing the necessity of establishing a mechanistic link between in vitro dissolution kinetics (e.g., in biorelevant media) and in vivo dissolution kinetics (e.g., in the human gastrointestinal tract). This includes determining which in vitro dissolution experiments are needed and how in vitro conditions can be translated to in vivo scenarios.
 - *Impact of formulation factors*: Considering how different formulation strategies (e.g., conventional or enabling formulations) can influence dissolution and absorption of the drug. Additionally, it may be crucial to understand the relationships among critical quality attributes (CQAs), critical material attributes (CMAs), critical process parameters (CPPs), and predefined clinical performance metrics (e.g., C_{max} and AUC).
- *Specifying dosing regimens and study conditions*: The simulations should mimic the clinical scenarios of interest. This includes the drug (and any interacting drugs), dose levels, dosing frequency, duration of treatment, and route of administration. For CoUs involving food effects, simulations would incorporate different meal types (e.g., standard high-fat, high-calorie meal; low-fat meal; fasted state) by adjusting relevant gastrointestinal physiological parameters within the PBPK model (e.g., gastric emptying time, pH, bile salt concentrations) [78, 79]. The OSP Suite's advanced absorption models facilitate such detailed simulations [47]. For DDI predictions, scenarios involve co-administration of the investigational drug (as victim or perpetrator) with relevant interacting drugs, using appropriate interaction models (e.g., competitive inhibition, mechanism-based inactivation, induction) built into PK-Sim/MoBi [47].
- *Defining simulation endpoints*: Clear, clinically relevant endpoints from the simulations should be predefined in the MAP to address the QoI. These could be specific pharmacokinetic parameters (e.g., AUC, C_{max} , C_{trough} , T_{max}), ratios of these parameters (e.g., AUC ratio for DDI assessment), target tissue concentrations, or the probability of achieving a certain exposure threshold. These endpoints often align with criteria used in clinical studies or those specified in regulatory guidance.

2.5.2 | Incorporating and Quantifying Uncertainty and Variability

A key strength of PBPK modeling is its ability to explore the impact of uncertainty and variability on pharmacokinetic outcomes.

- *Parameter uncertainty*: Uncertainty associated with key model input parameters (both drug-specific, like in vitro clearances or permeability, and system-specific, if customized) should be propagated through the model to understand its impact on the predictions relevant to the QoI. By assigning distributions to uncertain parameters, a distribution of predicted outcomes can be generated, providing prediction intervals (e.g., 5th-95th percentile) around the simulated mean or median PK profiles or parameters [97, 109].
- *Inter-individual variability (IIV)*: Accounting for inter-individual variability is crucial for the characterization of the expected range of PK responses and the identification of potential sources of high variability. Indeed, PBPK models can mechanistically account for sources of IIV by incorporating variability in physiological parameters (e.g., organ volumes, blood flows, enzyme/transporter expression levels) within the virtual population [46, 47]. To better define these parameters, new frameworks are being developed, such as coupling whole-body PBPK models with SAEM algorithms to estimate population parameters and inter-individual variability directly from individual concentration data [110]. Simulating a sufficiently large virtual population (e.g., 500–1000 subjects) is critical for applications like predicting the range of exposures or to identify sub populations at risk (under or over exposed). When assessing the probability of success of virtual bioequivalences with parallel design, simulating a minimum of 100 times the number of planned subjects in the trial is good practice.
- *Inter-occasion variability (IOV)*: Accounting for inter-occasion variability can also be crucial, in the context of bioequivalence studies, that are commonly performed as cross-over studies. IOV can either be applied to post hoc pharmacokinetic estimates, as it has been done in the context of DDI simulations [111], and obviously in the context of vBE [112] or IOV can be estimated from multidose studies, and included on relevant physiological parameters known to vary from one occasion to the other (e.g., gastric emptying). This is critical for virtual bioequivalence assessments aimed at for example probability of success and study design optimization (e.g., sample size).
- *Role of sensitivity analysis in application*: Sensitivity analyses, conducted during model evaluation, can also inform the application phase by highlighting which parameters or physiological processes most significantly drive the predicted outcomes for the specific QoI. This understanding helps in interpreting simulation results and focusing on the most critical uncertainties [75].

2.5.3 | Executing Simulations and Interpreting Results in Context

- *Simulation execution*: The number of virtual subjects or trials simulated should be adequate to provide robust

estimates for the endpoints of interest, especially when assessing variability or comparing different scenarios (e.g., DDI vs. no DDI, fed vs. fasted).

- *Examples of context-specific application points*: The application of a PBPK model is always driven by the CoU. For each application, a thorough model risk assessment—considering the model influence on decision-making, model impact, and the Consequence of a Wrong Decision [33]—is essential. This assessment dictates the necessary level of model credibility required, which is established through rigorous verification, validation against relevant data, and comprehensive uncertainty/sensitivity analyses.
 - *DDI Assessment (CoU-driven): Application*: Simulating co-administration to predict changes in exposure (AUC, $C_{\max}$ ratios) [113]. *Model risk considerations*: Incorrectly predicting the absence of a DDI could lead to patient safety issues if a clinically relevant interaction occurs; conversely, over-predicting an interaction might lead to unnecessary dose restrictions. A high-impact model that is also rated as highly influential (e.g., to waive a clinical DDI study) and has a high consequence of a wrong decision will necessitate high credibility. To assess the risk, a good understanding of exposure-safety and exposure-efficacy is further necessary. Therefore, the model evidence is not solely based on PBPK, but also knowledge of exposure-response and exposure-safety to assess the consequences of a wrong decision for patients. *Establishing credibility*: Crucial aspects include: accurate in vitro parameters for inhibition (K_i , IC_{50}) or induction (EC_{50} , $E_{\max}$) for relevant enzymes and transporters; mechanistically sound implementation of interaction dynamics (e.g., reversible, time-dependent inhibition; induction) within the PBPK platform; thorough model validation using observed clinical PK data for the investigational drug and for prototype substrates/inhibitors/inducers of the pathways involved, including demonstrating the model can recover known clinical DDI magnitudes; and sensitivity analysis to identify which parameters (e.g., f_m , CYP3A4, K_i) most impact DDI predictions.
 - *Special populations (CoU-driven)*:
 - *Pediatrics: Application*: Predicting PK across different pediatric age groups to inform dose selection, often aiming to match adult exposures or achieve a target therapeutic window [10]. *Model risk considerations*: Erroneous dose predictions in children carry significant safety (overdosing) or efficacy (underdosing) risks. *Establishing credibility*: Key elements include robust implementation and justification of age-dependent physiological scaling, including validated ontogeny functions for key metabolic enzymes and transporters and accounting for uncertainty if several different ontogeny functions exist; verification of adult model parameters before scaling; validation against available pediatric PK data whenever possible, even if sparse, across different age strata. If no pediatric data are available, extensive justification of the extrapolation approach and sensitivity to key assumptions (e.g., ontogeny profiles) is critical; in such cases, data from pharmacologically related compounds with the same mechanism of action and

established pediatric profiles may serve as a supportive basis for the proposed model.

- *Organ impairment (renal or hepatic): Application:* Predicting exposure changes in patients with varying degrees of renal or hepatic impairment to guide dosing adjustments. *Model risk considerations:* Incorrect dose adjustments can lead to toxicity or loss of efficacy. *Establishing credibility:* Requires accurate PBPK model representation of how the specific organ impairment impacts drug ADME (e.g., changes in GFR for renal impairment; alterations in liver blood flow, changes in expression/function of transporters and enzymes, protein binding for hepatic impairment), and furthermore, it is important to consider that uremic toxins from renal impairment often impact the function of hepatic transporters and metabolizing enzymes; validation of the base model in healthy subjects; validation against clinical PK data from subjects with relevant degrees of organ impairment if available. If not, justification for the extrapolation based on known drug properties and disease impact on relevant physiological pathways is crucial, supported by sensitivity analyses.
- *Formulation changes/food effects (CoU-driven): Application:* Predicting relative bioavailability of new formulations, assessing vBE, or quantifying the impact of food on drug absorption [74]. *Model risk considerations:* Incorrect predictions could ultimately harm patients by promoting a sub-optimal formulation that compromises safety or efficacy, or by unethically exposing volunteers to a drug in a futile bioequivalence study. *Establishing credibility:* Depends on: accurate in vitro characterization of drug substance (e.g., solubility in biorelevant media) and formulation properties (e.g., dissolution profiles); robust mechanistic absorption models linking drug/formulation properties to in vivo release and absorption; for food effect models, appropriate parameterization of fed state GI physiology [78, 79, 97]; validation of the absorption model using clinical PK data from different formulations or under fasted/fed conditions. For vBE, depending on the study design demonstrating accurate reproduction of inter-individual variability (parallel study design) or inter-occasion variability (cross-over study design) is also important.
- *First-in-human (FIH) dose projections (CoU-driven): Application:* Predicting human PK profiles and variability from preclinical data to select a safe starting dose and inform the FIH trial dose range [21]. *Model risk considerations:* Significant misprediction could lead to an unsafe or sub-pharmacological starting dose (a concern for FIH studies using patients in ex. oncology). *Establishing credibility:* Relies on: quality and relevance of preclinical PK data (multiple species, IV data); robustness of interspecies scaling methods (IVIVE integrated into PBPK); thorough validation of animal PBPK models; comprehensive sensitivity analysis of influential parameters and assumptions (e.g., human CL_{int} prediction, K_p values, absorption rate) and their impact on predicted human PK. Uncertainty assessment is critical for defining a safe FIH dose range.
- *Biologics (e.g., monoclonal antibodies) (CoU-driven): Application:* Predicting human PK of monoclonal

antibodies (mAbs) from reference Kd_{FCRN} [114], understanding target-mediated drug disposition (TMDD), and exploring factors influencing mAb exposure and response. *Model risk considerations:* Mischaracterizing mAb PK, particularly non-linear aspects due to TMDD, can impact dose selection and efficacy/safety predictions. *Establishing credibility:* Requires appropriate representation of mAb-specific mechanisms (e.g., binding to soluble/membrane targets, neonatal Fc receptor (FcRn) recycling, lymphatic uptake) [115].

Table 1 provides a consolidated overview of the specific PBPK application points detailed above, highlighting the interplay between clinical risk and the evidence required for regulatory acceptance.

2.6 | Documentation for Regulatory Interaction

Comprehensive, transparent, and well-organized documentation is a critical prerequisite for the regulatory acceptance and efficient review of any MIDD submission involving PBPK models [16, 33, 37]. For regulatory interactions the fundamental goal is to provide a clear and complete understanding of the modeling approach, underlying data and assumptions, the evaluation performed, and the rationale for concluding that the model is fit for its intended CoU in addressing the specific QoI [33].

To achieve this, the entire application of the PBPK model—from simulation plan to final interpretation—must be documented in a comprehensive analysis plan and analysis report. This report is the primary vehicle for demonstrating the model's credibility and must transparently include:

- A clear linkage of the simulation design to the specific CoU and QoI.
- A detailed description of the virtual populations, dosing regimens, and scenarios simulated.
- A full presentation of simulation results, including graphical and quantitative summaries.
- Complete disclosure of the uncertainty and variability assessment
- A thorough discussion of the findings, their implications for the QoI, and how they support the proposed decision-making while explicitly acknowledging all model limitations.

2.6.1 | General Principles for MIDD Documentation

General principles for MIDD documentation are comprehensively defined in regulatory guidelines [16, 33, 37] and industry good practice paper [42]. Key requirements include:

- *Prospective planning:* Utilizing a Model Analysis Plan (MAP) that details the QoI, CoU, proposed strategy, and predefined evaluation criteria [33, 42].

TABLE 1 | Relationship between context of use (CoU), model risk, and credibility requirements for PBPK applications.

Application (CoU)	Primary goal	Model risk considerations	Key credibility factors	Regulatory outcome
DDI assessment	Predict changes in exposure (AUC, C_{max}) during co-administration	Safety risks if interaction is missed or underestimated; unnecessary restrictions if over-predicted	Accurate $K_i/IC_{50}/E_{max}$; mechanistic interaction dynamics; recovery of known clinical DDI magnitudes	Waiving clinical DDI studies; informing drug-drug interaction sections of the product label
Special populations				
Pediatrics	Inform dose selection to match adult exposure or therapeutic windows	Significant safety (overdosing) or efficacy (underdosing) risks in children	Validated ontogeny functions (metabolism/transporters); age-dependent physiological scaling; sparse data validation	Dose selection for pediatric trials; support for pediatric labelling when data is limited
Organ impairment	Guide dosing adjustments for renal or hepatic impairment	Toxicity or loss of efficacy due to incorrect adjustment	Accurate representation of ADME changes (GFR, liver blood flow, expression/function of transporters and enzymes); validation against healthy and impaired PK data	Dose adjustment recommendations in the “Special Populations” section of the label
Formulation & food effects	Predict bioavailability, assess virtual BE (vBE), or quantify food impact	Promoting sub-optimal formulations; unethical exposure in futile bioequivalence studies	In vitro solubility/dissolution; mechanistic absorption models; parameterization of fed-state GI physiology	Generating evidence for bioequivalence via virtual bioequivalence simulation; support for biowaivers; determining food effect labelling; bridging formulation changes
FIH dose projections	Predict human PK from preclinical data to select safe starting doses	Unsafe or sub-pharmacological starting doses, particularly in oncology patients	Quality of multi-species data; IVIVE integrated into PBPK; animal model validation; uncertainty/sensitivity analysis	Approval of FIH trial protocols; defining safe starting doses and escalation increments
Biologics (mAbs)	Understand TMDD and factors influencing mAb exposure/response	Mischaracterizing non-linear PK impacts efficacy and safety predictions	Representation of mAb mechanisms (FcRn recycling, lymphatic uptake, target binding/TMDD)	Informing dose-exposure-response; characterization of non-linear clearance for biologics

- *Transparent reporting*: Generating a comprehensive Model Analysis Report (MAR) documenting model structure, assumptions, input data, evaluation outcomes, and limitations [33, 42].
- *Traceability and reproducibility*: Providing sufficient methodological detail, including software versions, electronic model files, and datasets, to allow independent replication by regulatory agencies [33, 42]. Therefore reproducibility, automation, traceability, and standardization are considered key aspects of any analysis report [116].

2.6.2 | PBPK-Specific Documentation for Regulatory Review

Building on the general MIDD documentation principles, PBPK-specific documentation should also provide detailed aspects unique to its methodology [16, 34, 37]:

- *Model structure and system parameters*: Clear description and visual representation (schematic) of the PBPK model structure (organs, compartments, interconnections, physiological equations if non-standard). Detailed sourcing and justification for all system parameters (organ volumes, blood flows, tissue composition), including population data characteristics and any modifications from defaults. PBPK software platform and version.
- *PK experimental and study data*: Detailed schematic representation or list, in addition to the data summary table, representing which data informs what
- *Drug-specific input parameters*: Exhaustive list of all drug input parameters (physicochemical properties, in vitro ADME data like metabolic CL_{int} values, transporter kinetics) with values, units, and precise, justified sources (study reports, literature, in silico tools). Detail IVIVE scaling methods. Robust justification for parameter selection, especially if conflicting data exist or values were optimized.
- *Software and critical model settings*: Beyond platform/version, report specific software choices impacting outcomes (algorithms for K_p /permeability, absorption/dissolution models, critical non-default solver settings)
- *Model development narrative*: Transparent, step-by-step description of model construction and refinement, rationale for structural choices. Describe sequential IV-then-PO or simultaneous fitting approaches and justification.
- *Model verification documentation*: Evidence of technical soundness (e.g., mass balance checks, numerical stability, unit consistency). For custom code or significant modifications, document more extensive verification.
- *Model validation and applicability*: Clear documentation of assumptions and approximations. In the analysis plan, prospectively define acceptance criteria and present results of all model evaluation exercises (graphical comparisons, quantitative goodness-of-fit metrics, consistency checks, sensitivity/uncertainty/scenario analyses) and, importantly,

the rationale for deeming the model qualified for its CoU in the report.

- *Model files and datasets*: Regulatory agencies require submission of actual PBPK model files and key datasets to facilitate review and ensure reproducibility [16, 37]. The Model Master File (MMF) concept is an emerging framework to streamline submission, review, and sharing of well-validated models and modeling platforms [117, 118]. An MMF can serve as a comprehensive repository containing detailed information on a model's development, extensive verification and validation evidence, and its qualified CoU(s).

Adherence to these documentation principles facilitates a more efficient, transparent, and robust regulatory assessment, ultimately contributing to greater confidence in PBPK modeling.

3 | Conclusion

This paper presents a best-practice PBPK focused framework, detailing a structured process for robust model development, evaluation, and reporting aligned with regulatory expectations.

This process commences with prospectively defining the QoI, CoU, and associated risks (Model Influence, Consequence of a Wrong Decision) per ICH M15. This definition dictates the necessary rigor for all subsequent activities. In addition, it is highly advisable to make use of the “assessment table” provided in the ICH M15 guideline in order not only to aid in regulatory interaction but also in discussions proceeding regulatory interaction. The central tenet is the ‘fit-for-purpose’ principle: a model's credibility must be demonstrably sufficient for its CoU, proportional to the influence and risk of the decision it supports.

Adhering to these practices—including systematic development, formal evaluation, and comprehensive documentation—will enhance the quality and regulatory acceptance of PBPK submissions, solidifying their role in quantitative decision-making.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** ICH M15 “assessment table”—Vericiguat Case Example.